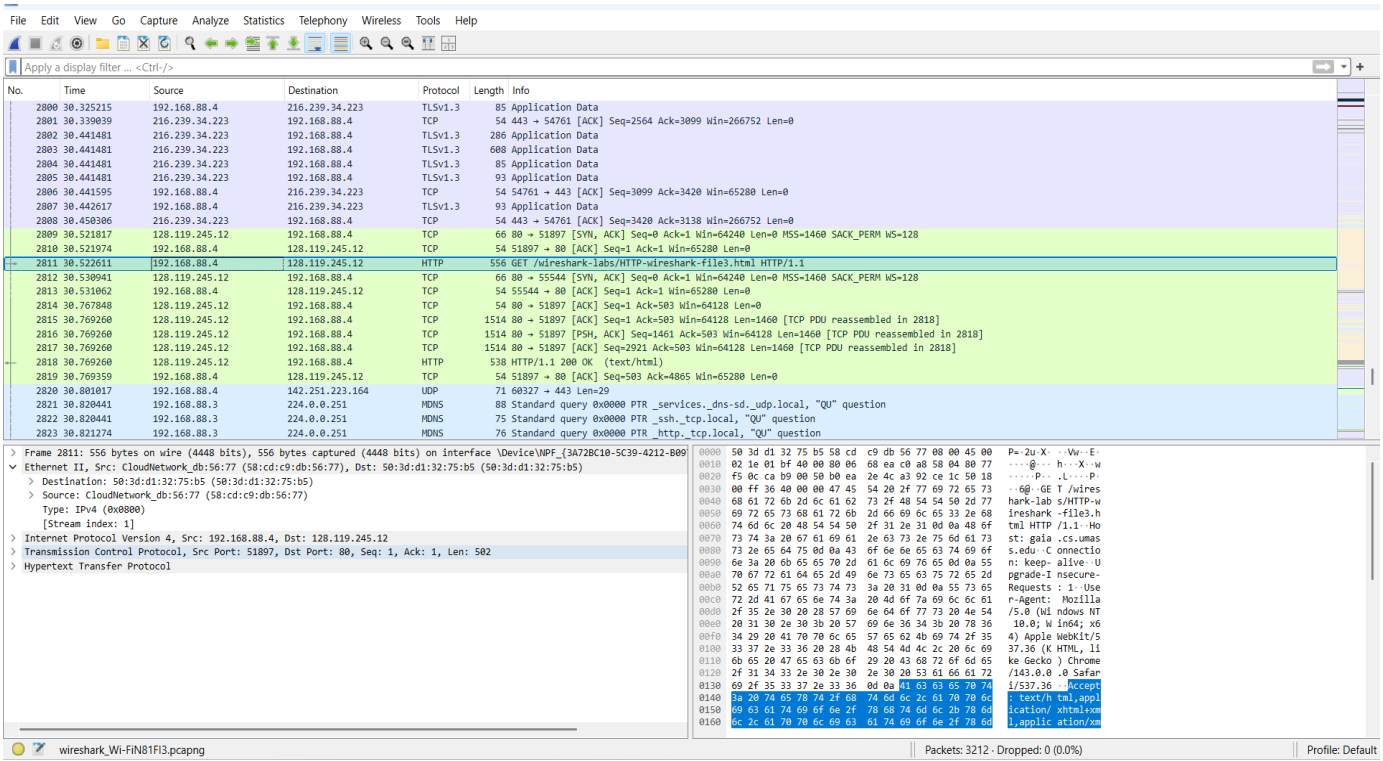
COMPUTER NETWORKS - UE24CS252B

Lab 5: Understanding Ethernet and ARP using Wireshark

The objective of this experiment is to study the **Ethernet protocol and the Address Resolution Protocol (ARP)** using Wireshark. The experiment involves capturing and analyzing Ethernet frames to understand their structure and examining the process by which ARP resolves IP addresses to corresponding MAC addresses within a local network.

PART A

1. Start Wireshark and enter the following URL into your browser: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file3.html.
2. Stop Wireshark packet capture.
3. In Wireshark apply the **http** filter to get the packet which sent the **GET** request to the server. [You may have to clear the cache in your browser if you are not able to find this packet. On Chrome Ctrl+Shift+Delete]



Deliverables for PART A

1) Provide a **screenshot of the packet that contains the HTTP GET request sent to the server**, including the **Ethernet frame information**.

2) In this packet, **what is the Ethernet source address?** Is this the **physical (MAC) address of your device**?

3) In the same packet, **what is the Ethernet destination address?** Is this the **physical address of the server?** If not, provide a justification.

4) What is the **destination IP address** in this packet? Does it correspond to the **IP address of the server**?

5) Provide a **screenshot of the packet containing the HTTP text response from the server**, including the **Ethernet frame information**.

6) Compare this packet with the earlier HTTP GET packet. **Explain the changes observed in the Ethernet source and destination addresses**, and identify **which devices these addresses belong to**.

PART B

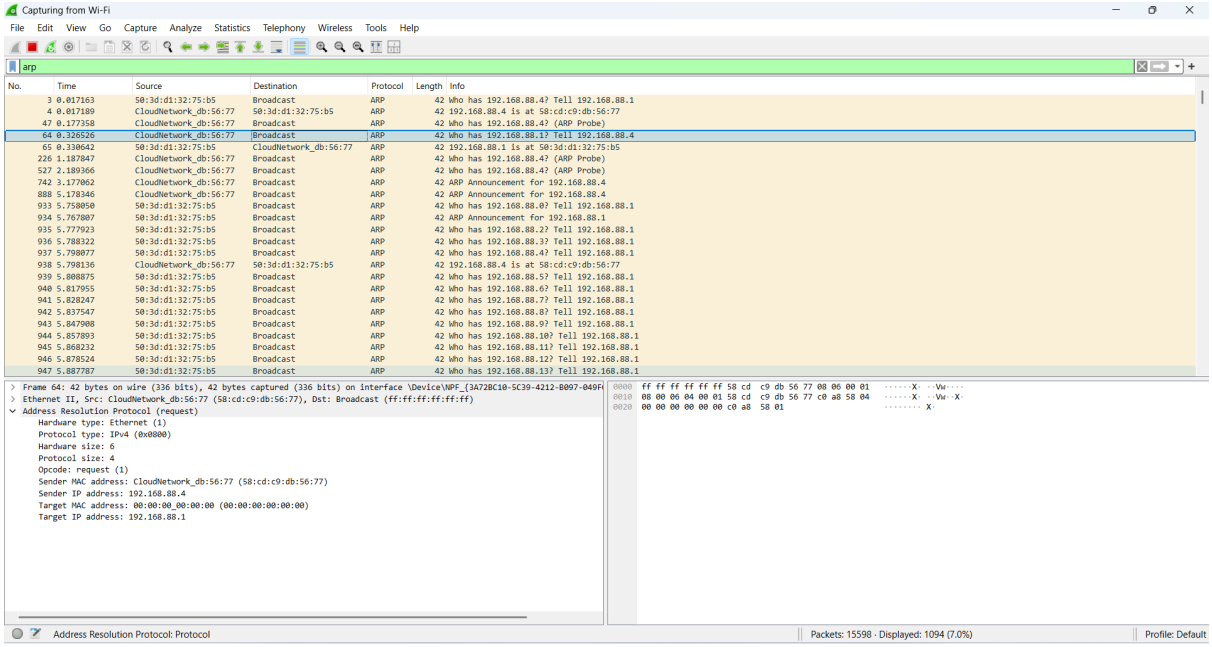
In this part, we aim to observe the ARP packets transmitted in the network. Before beginning the experiment, the existing ARP table on the device should be cleared.

Follow the below steps below to clear the ARP table and see the ARP entries:

1. Disconnect from the network you are connected to currently
2. On windows , open command prompt in administrator mode and enter
   i. On Windows: arp -d *

ii. On Mac: - sudo arp -a -d
(which will delete the existing arp table entries)

3. Open wireshark and start the packet capture and enter **arp** as the filter
4. Connect back to the network
5. Once you are connected back to the network, you should be able to see the arp packets being exchanged between your device , other devices in network and the gateway of your network



Deliverables for PART B

1. Provide a **screenshot of the ARP probe packet**, clearly showing the **sender IP address and target IP address**. Explain **why this packet is transmitted**.

2. Provide a **screenshot of the ARP packet sent from your device**, including the **sender IP address and target IP address**.

   A. What differences do you observe when compared with the **ARP probe packet**?
   B. What is the **target MAC address** in this packet? Justify your answer.

C. Which **opcode value** in the packet indicates that it is an **ARP request**?

3. Provide a **screenshot of the ARP response packet** received after the ARP request, showing the **sender and target IP addresses**.

   A. What changes do you observe compared to the ARP request packet?
   B. What is the **target MAC address** in this packet? Justify your answer.
   C. Which **opcode value** indicates that the packet is an **ARP reply**?